

Group 22

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The Cosmic Microwave Background Radiation

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WHAT THIS PRESENTATION CONTAINS

- What is CMB and how it was made.
- How CMB was discovered.
- The various space missions that have studied CMB.
- Topology and composition of the Universe.
- Anisotropies in CMB.
- How CMB changed the value for Hubble's constant.
- The future of CMB in research and theories.



WHAT IS THE COSMIC MICROWAVE BACKGROUND?

- CMB is the radiation leftover from the creation of the Universe.
- It is in the microwave region.
- The image of CMB consists of various hot and cold regions, with the bluer regions being colder and the redder regions being hotter. (In some maps, red represents cold, and blue represents hot).

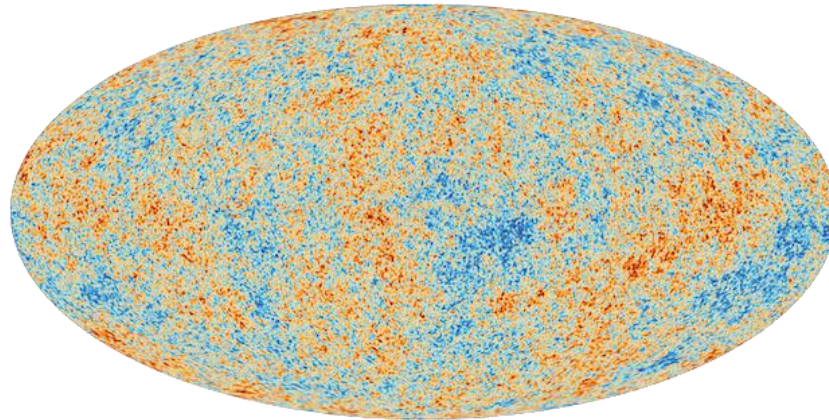


Image from ESA

https://www.esa.int/ESA_Multimedia/Images/2013/03/Planck_CMB



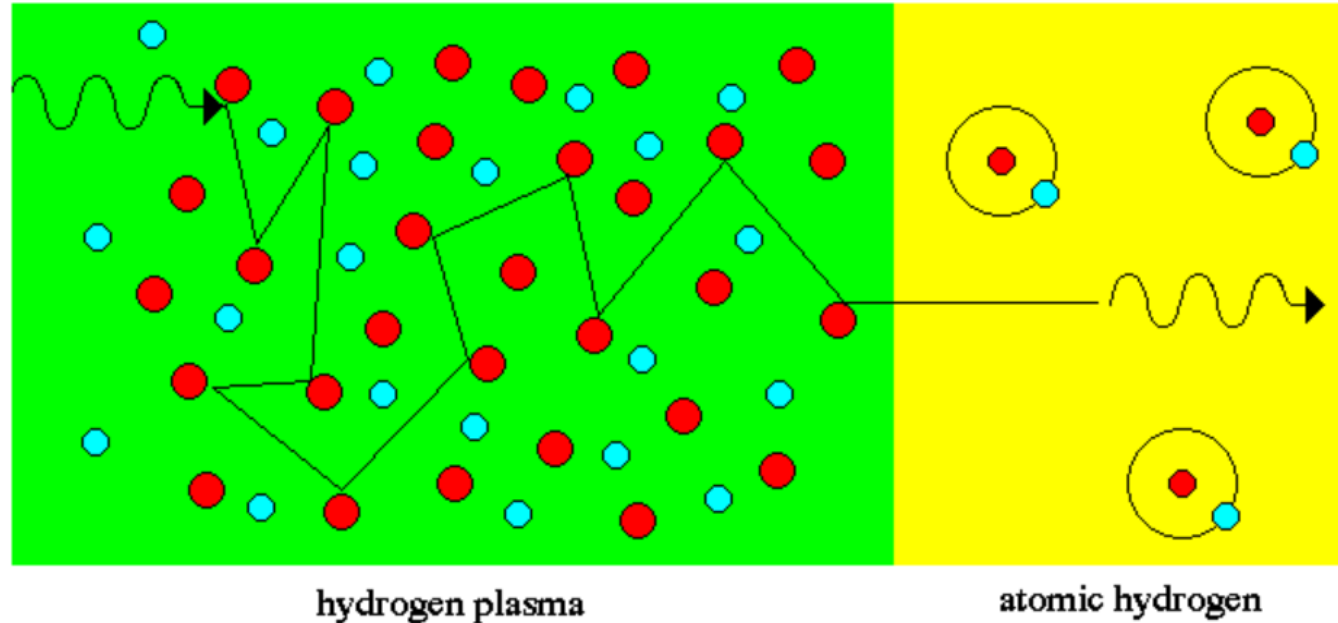
HOW WAS CMB CREATED?

- 13.8 billion years ago a few moments after the Big Bang the universe was a concoction of particles which collided and interacted with each other.
- After 380,000 years the universe expanded and cooled which caused recombination.
- Protons and electrons combined to form neutral hydrogen atoms.
- This caused free photons to move in all directions emitting radiation.
- The leftover radiation is what we detect today.



Last Scattering Epoch

As the Universe cooled, the free electrons and protons could finally bond together to form hydrogen atoms. At the same time, the Universe went from a rich plasma to a gas of neutral hydrogen.



In a plasma, the mean free path of a photon is very short. In a gas of atomic hydrogen, the mean free path is very long, as long as the size of the Universe. Thus, the transition from the early plasma to atomic hydrogen is the epoch of last scattering, the point in time when the photons became free to travel without hindrance.

<https://pages.uoregon.edu/jschombe/cosmo/lectures/lec23.html>



HOW WAS CMB DISCOVERED?

- Discovered accidentally in 1965 by Arno Penzias and Robert Wilson.
- Used horn reflector antenna in Bell laboratories in New Jersey when they stumbled upon a very persistent background noise.
- After eliminating all possible sources of interference and error they found a faint glow of radiation coming from all directions of space.
- This was CMB it has red-shifted to the microwave region of EM spectrum, and it holds very subtle temperature fluctuations.
- This discovery led to Penzias and Wilson winning the Nobel Prize for Physics in 1978.



COSMIC MICROWAVE BACKGROUND MISSIONS



COSMIC BACKGROUND EXPLORER (COBE)

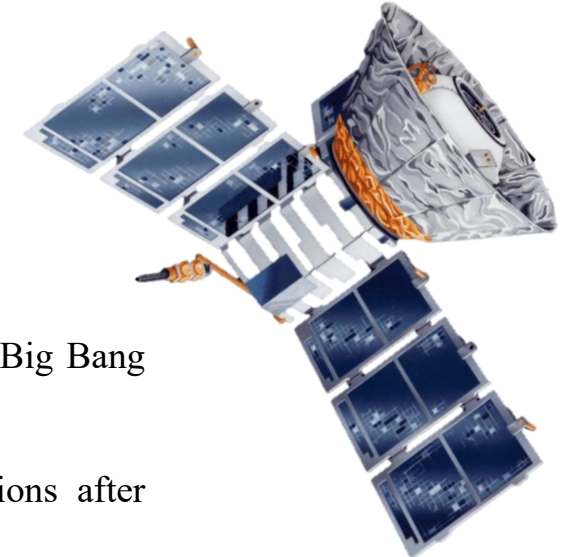
AIM

1. Launched in 1989, COBE's primary goal was to measure the diffuse infrared and microwave radiation from the early universe with a much higher precision than previous experiments.
2. Equipped with relatively basic instruments like the FIRAS and the DMR, considered groundbreaking at the time.

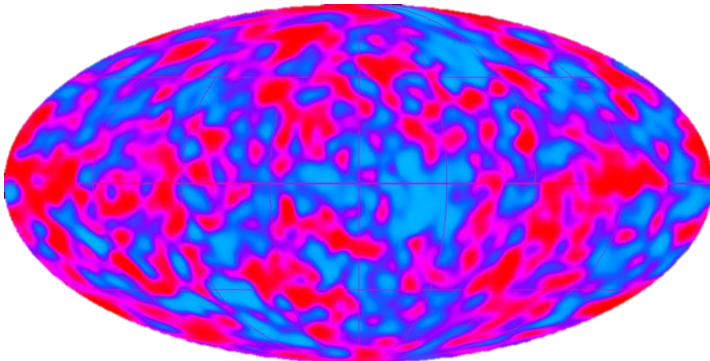
ANALYSIS

Made 3 major discoveries

1. Providing precise measurements of the CMB spectrum, confirming that it did indeed match the predictions of the Big Bang Theory nearly flawlessly.
2. Showed temperature isotropies across the heat map, which would later show tiny anisotropies in future missions after improving on resolution and precision.
3. Confirmed the blackbody spectrum of the CMB and detected large-scale anisotropies.



https://commons.wikimedia.org/wiki/File:Cosmic_Background_Explorer_spacecraft_model.png



IMPACT

1. These findings had a monumental impact; confirmation of the Big Bang theory earned John C. Mather and George F. Smoot the Nobel Prize in Physics, 2006. It would go on to lay the groundwork for further experiments.
2. The Big Bang Theory and The Steady State Model of the universe were one of the most heavily debated topics in cosmology, the importance of confirmation of the Big Bang is often underestimated! It would save years of experiments to settle this debate.



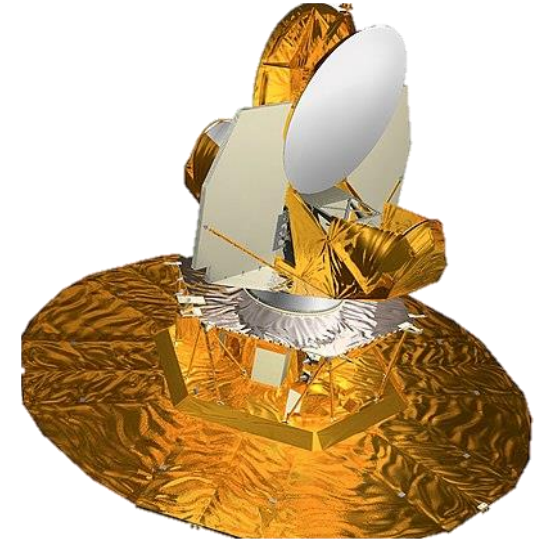
WILKINSON MICROWAVE ANISOTROPY PROBE (WMAP)

AIM

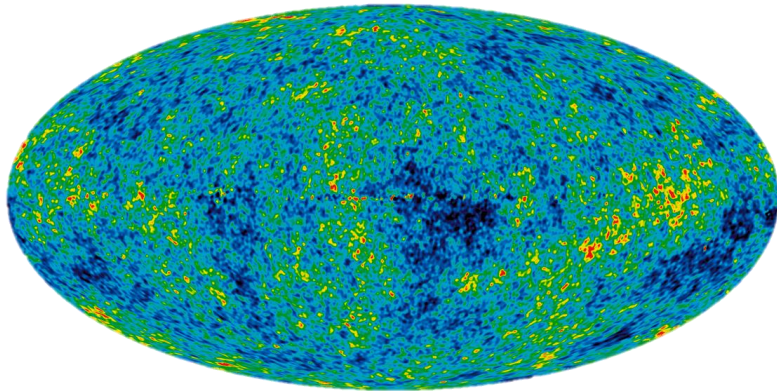
1. Successor to the COBE, it was launched in 2001 with the aim of improving on the resolution of the temperature anisotropy maps in the CMB.

ANALYSIS

1. Provided a full-sky coverage with a resolution **35** times finer than COBE.
2. The high-resolution data allowed for a more detailed study of these anisotropies, they led to precise measurements of several key cosmological parameters, including the age of the universe, the density of atoms, the density of all other matter, the epoch of reionization, and the curvature of space.



https://en.wikipedia.org/wiki/File:WMAP_spacecraft.jpg



<https://www.astronomy.com/wp-content/uploads/sites/2/2023/02/CM3.jpg?resize=477%2C412>

IMPACT

1. Provided strong evidence for dark energy and helped constrain its properties, a highly sought-after piece of the puzzle in cosmology, and the early universe.
2. Results from WMAP led to the establishment of the Λ CDM (Lambda-Cold Dark Matter) model as the standard model of cosmology!



PLANCK MISSION AND SPACECRAFT

AIM

1. Launched in 2009, it aimed to analyse the Cosmic Microwave Background with greater sensitivity and angular resolution than COBE and WMAP. It was concerned with offering deeper insights into the cosmological parameters and the standard model of cosmology.

ANALYSIS

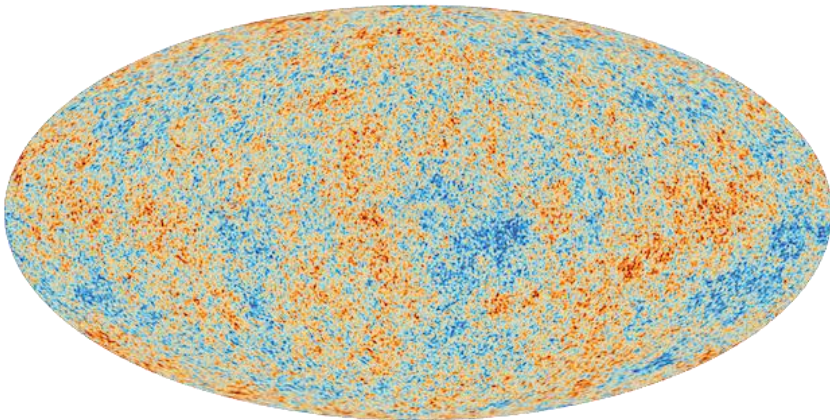
1. Planck had a higher sensitivity and angular resolution, with two instruments; the Low Frequency Instrument (LFI) and the High Frequency Instrument (HFI).
2. Increasing precision of the measurements led to tighter constraints on cosmological models.



https://www.cam.ac.uk/sites/www.cam.ac.uk/files/inner-images/planck_satellite.jpg

IMPACT

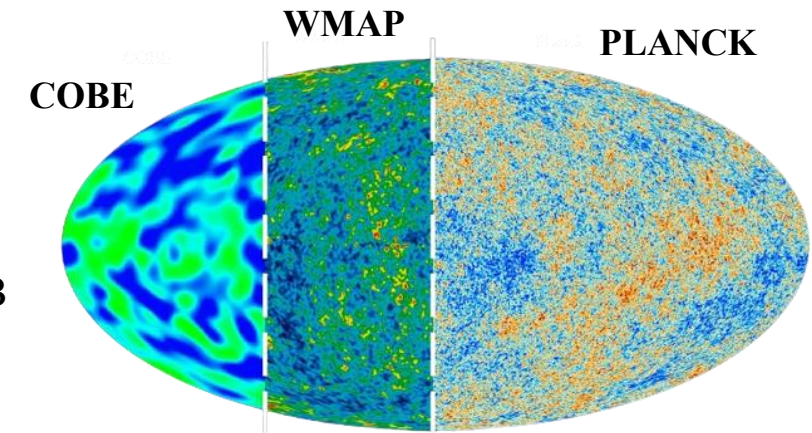
1. Refined the values for Hubble constant and the age of the universe.
2. Results from this mission offered a new perspective on anomalies in the CMB, including the ‘cold spot’ and other features not predicted by the standard model.



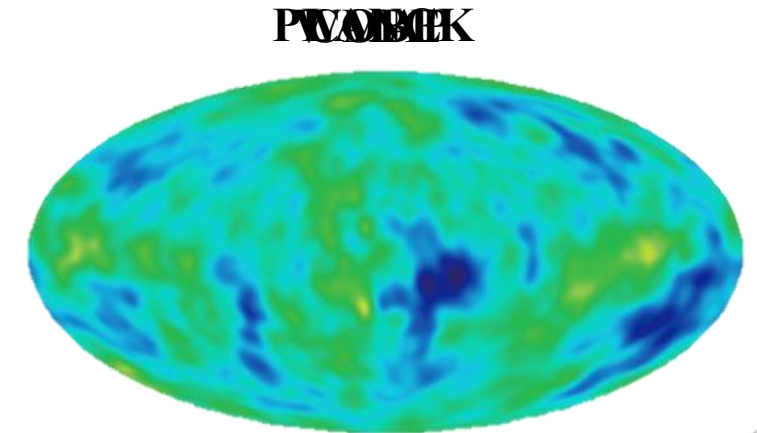
COBE vs WMAP vs PLANCK

A COMPARISON

- Each mission represents a step forward in technology, data analysis, and theoretical understanding.
- COBE showed the CMB having temperature isotropy, while WMAP would show the CMB with a significantly better resolution, showing very small *anisotropies* in temperature as well. Planck would further improve on this resolution to properly study these anisotropies, testing theoretical models for the universe, exploring their relation to the overall structure of the universe etc.
- Understanding their differences helps in comprehending how scientific instrumentation and methodology evolve over time in response to new challenges and questions.
- COBE, WMAP, and Planck each played an integral role in advancing our understanding of the universe. Their progression from COBE's initial large-scale observations to Planck's high-resolution mapping reflects both technological advancements and the evolving nature of cosmological questions. For physics students, these missions are exemplary in showcasing how empirical evidence can shape and refine theoretical models in science.



<https://imgur.com/eVapPtW>



THE HOLMDEL HORN ANTENNA

AIM

1. Built by Bell Labs, designed to be used with and early satellite, Echo. The goal was to collect, amplify and emit radio signals across large distances (for commercial use). After Echo became obsolete in just a few years, the antenna was free to be used for research.
2. Penzias and Wilson took this opportunity to use it for analysing radio signals collected from the spaces between galaxies.

ANALYSIS

1. Shortly after the antenna was employed for research, they encountered a persistent background ‘noise’, from all directions.
2. The two researchers would make numerous attempts to remove this background noise. One such way was purifying a nest of pigeons with the use of a shotgun. "And so the pigeons left with a smaller bang, but the noise remained, coming from every direction.“ - Penzias.
3. After consulting colleagues regarding this background noise, they would discover Robert H. Dicke, who realised that this was indeed the ‘relic radiation’ he theorised years ago. This would be the discovery of the Cosmic Microwave Background.



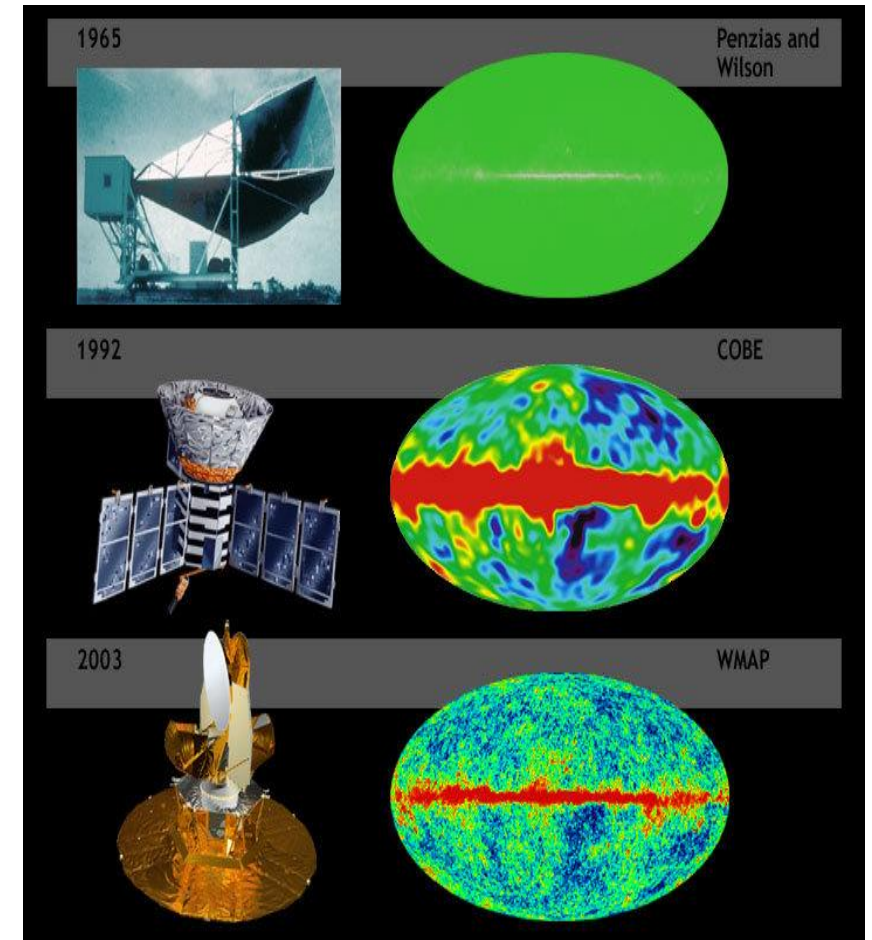
COMPARISON WITH HORN ANTENNA

1. TECHNOLOGY AND SENSITIVITY

- **Horn Antenna:** A basic radio antenna used in 1965, suitable for detecting microwave radiation.
- **COBE, WMAP, Planck:** These missions employed advanced satellite-based Instruments with far greater sensitivity and resolution, specifically designed for detailed CMB observation.

2. PURPOSE AND DESIGN

- **Horn Antenna:** Not originally designed for cosmological observations; its discovery of the CMB was unintentional.
- **COBE, WMAP, Planck:** Specifically designed for precise CMB observations and cosmological research.



COSMIC MICROWAVE BACKGROUND ANISOTROPIES

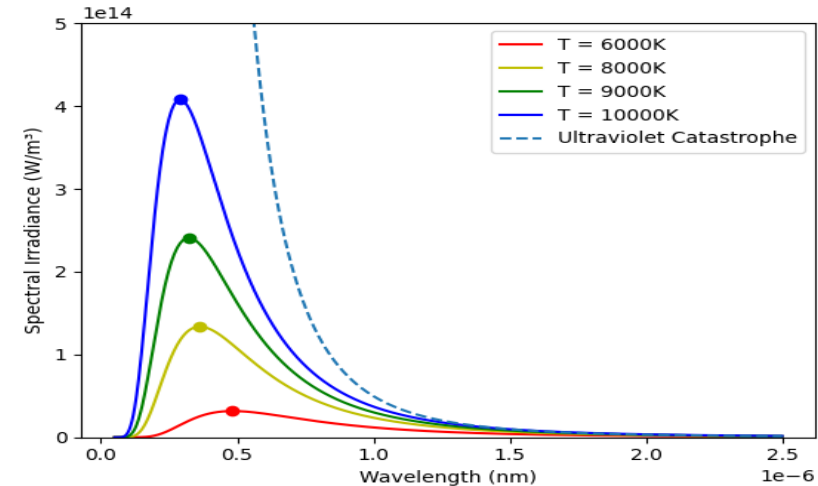
ISOTROPY

ANISOTROPY



CMB – AN EVIDENCE OF THE BIG BANG

- An ideal blackbody absorbs and emits radiation at the same rate.
- It absorbs and emits at all wavelengths, including visible.
- The CMB spectrum shows near-perfect conformity to the spectrum of an ideal blackbody.
- CMB behaving like an ideal blackbody means that the universe was once opaque.



Ideal blackbody curves at varying temperatures (Produced by Areeb)

ISOTHERMAL
OPAQUE
NON-REFLECTING

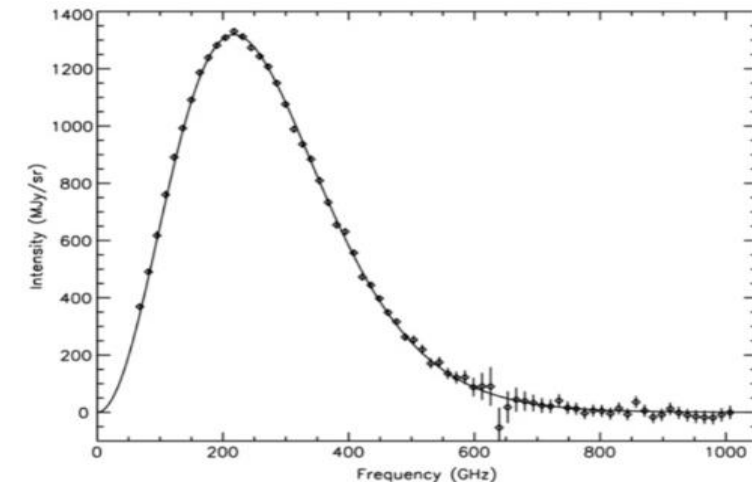


Figure 1. Blackbody spectrum of CMB as established by the FIRAS experiment on board *COBE* with WMAP recalibration (Fixsen 2009). (Courtesy of NASA.)

<https://www.sciencedirect.com/science/article/pii/S1355219816302039>

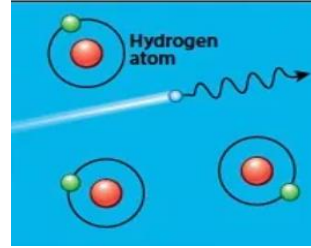


2 & 3) Recombination and Photon Decoupling

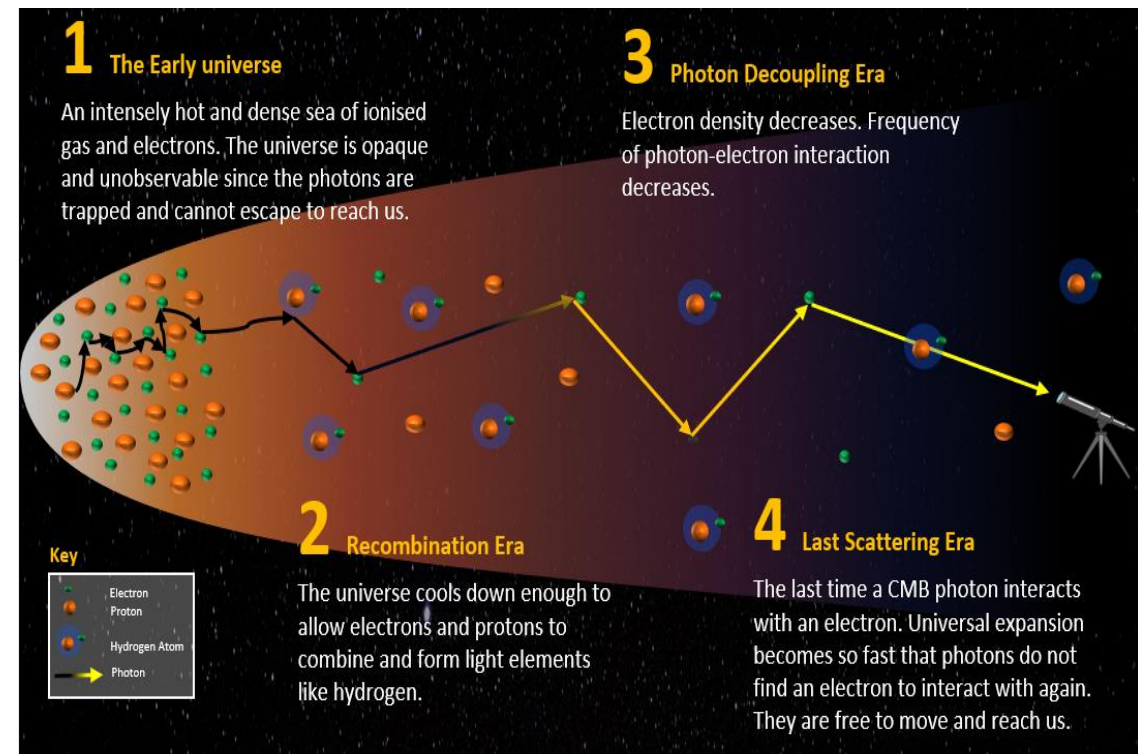
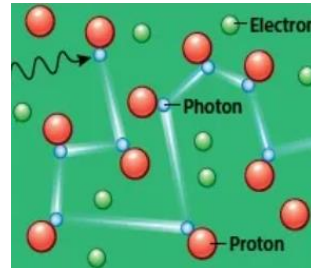
- The electron density in the universe has decreased.
- The photons pass through and do not scatter off of neutral atoms.

1) The Early Universe

- This necessitates a high density of electrons in earlier times for photon to scatter off.
- Neutral atoms existed as ionised gas, implying a very high initial temperature.



BACK
IN
TIME
↓



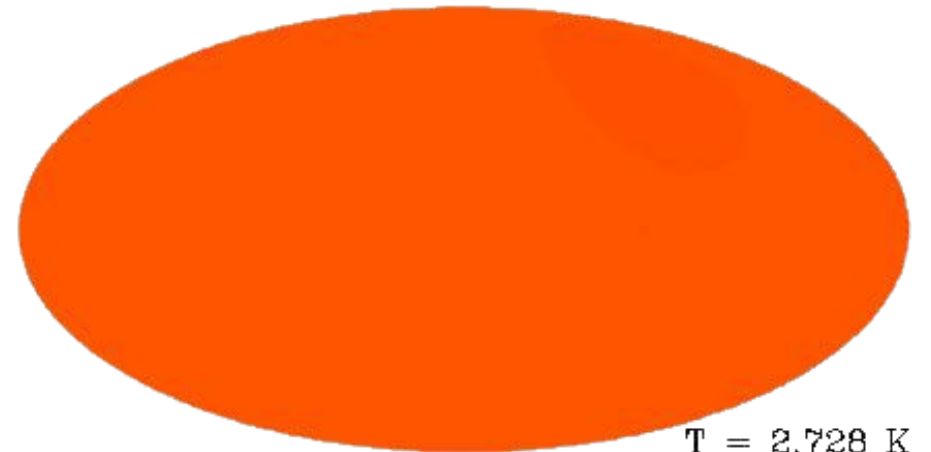
Epochs in the early Universe
Image Produced by Areeb

The isotropy of CMB temperature shows that all regions of the universe were close enough to exchange energy.

All this evidence supports the Big Bang Model.

Image taken from:

<https://lambda.gsfc.nasa.gov/product/suborbit/POLAR/cmb.physics.wisc.edu/polar/ezexp.html>

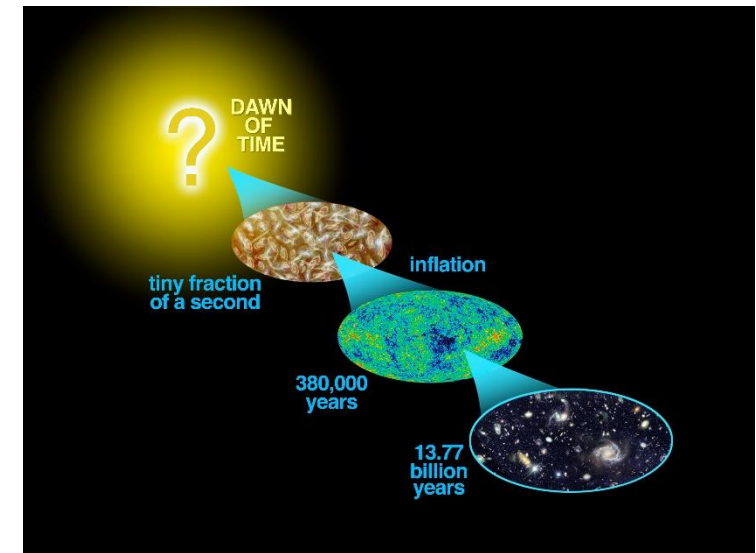
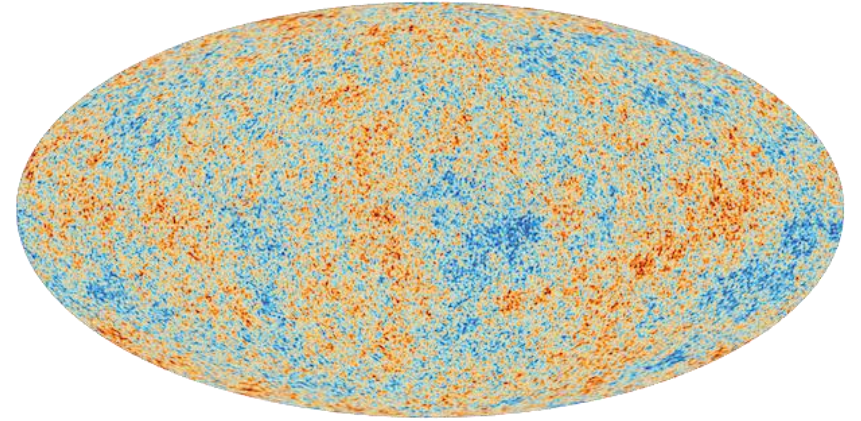


$T = 2.728 \text{ K}$



CMB TEMPERATURE ANISOTROPIES

- **Anisotropy:** Any physical property that is non-uniform in different directions.
- We see fluctuations of $20\ \mu\text{K}$ in CMB temperatures.
- These are a result of quantum fluctuations becoming inflated due to cosmic expansion.
- The fluctuations in temperature translate to fluctuations in density distribution.
- These fluctuations become precursors to the large-scale structures in the cosmos.

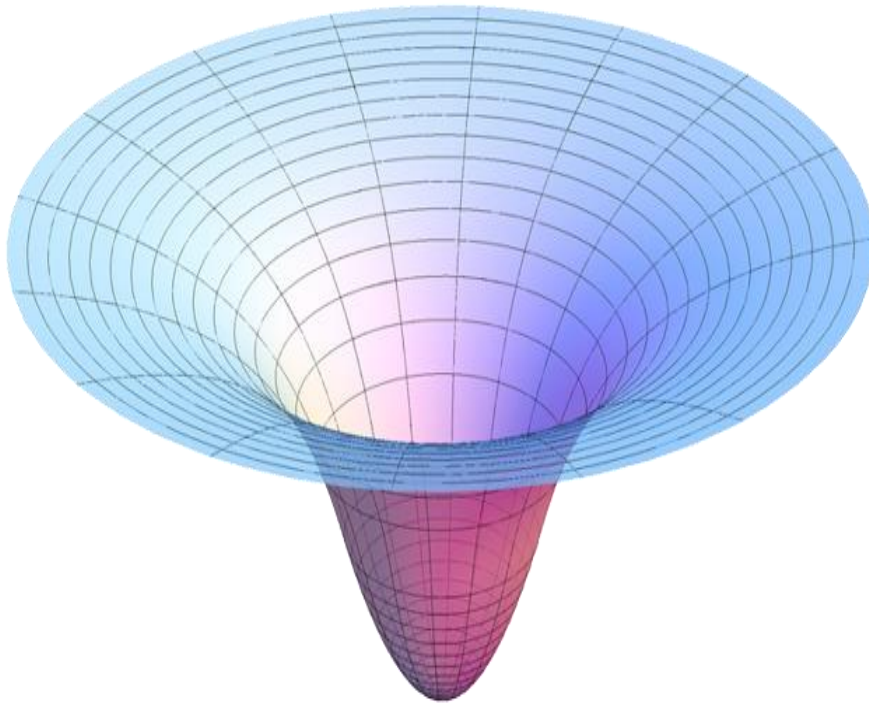


<https://map.gsfc.nasa.gov/media/020622/index.html>



An overdense region exerts a strong gravitational attraction visualised by a deep gravitational potential well

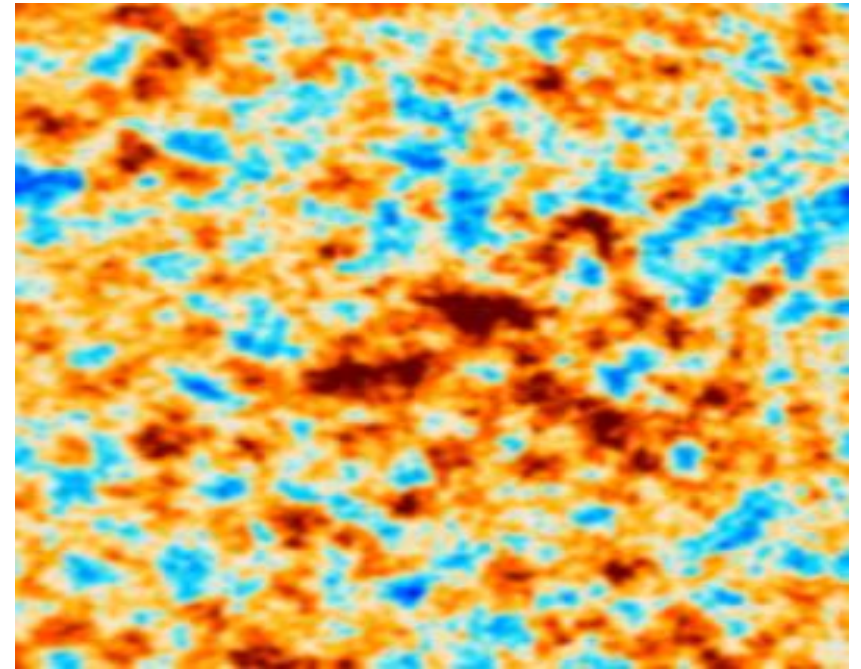
A photon trying to escape a deep well loses energy



<https://commons.wikimedia.org/wiki/File:GravityPotential.jpg>



Region becomes cooler
represented by a **red** spot on the
CMB map.
(in some colour scales, **blue**)



https://abaleato.github.io/cmb_visualisations/cmb.html

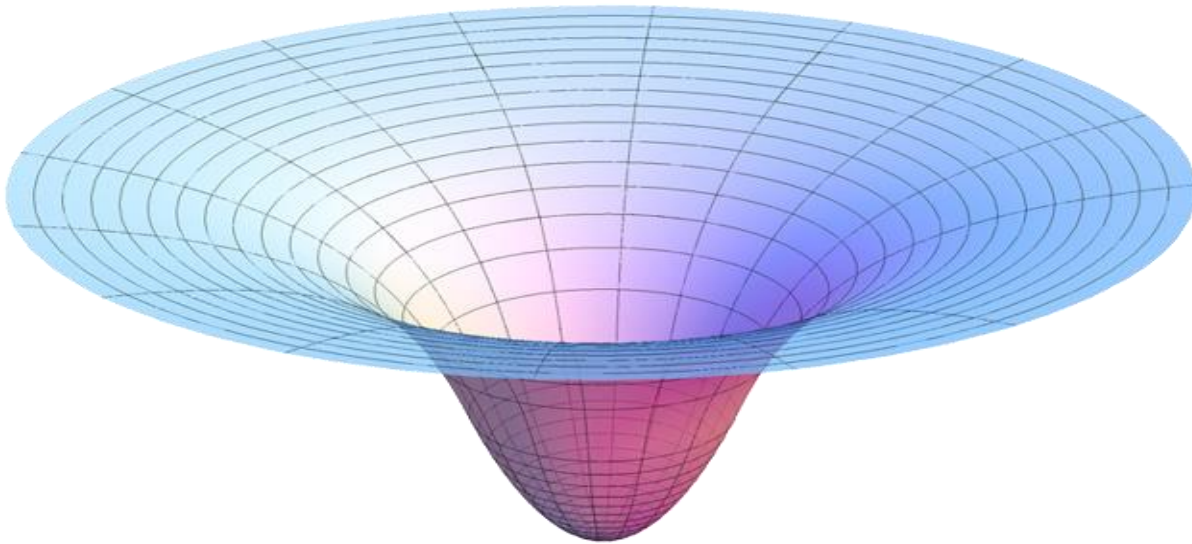


An underdense region exerts a weak gravitational attraction visualised by a shallow gravitational potential well

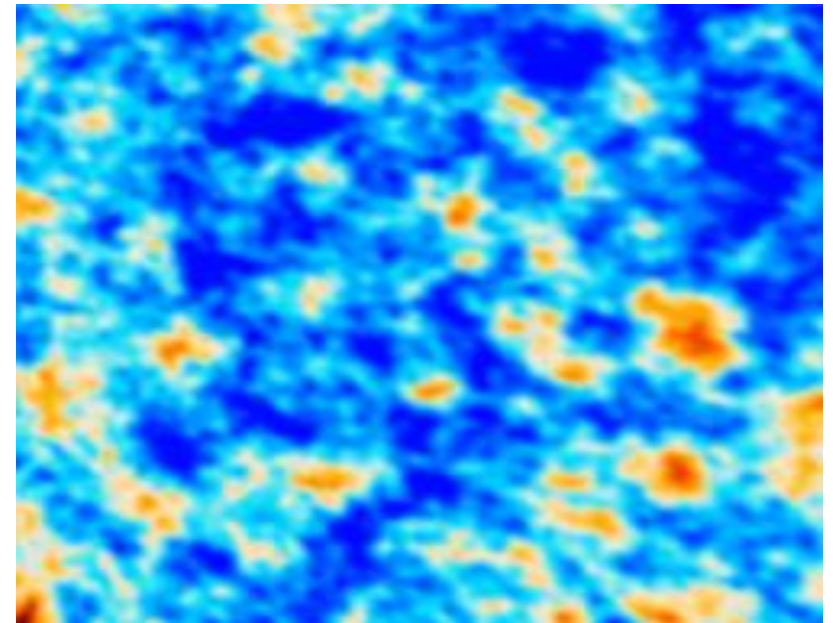
A photon trying to escape a shallow well retains much of its energy



Region is hotter represented by a blue spot on the CMB map.
(in some colour scales, red)



<https://commons.wikimedia.org/wiki/File:GravityPotential.jpg>



https://abaleato.github.io/cmb_visualisations/cmb.html

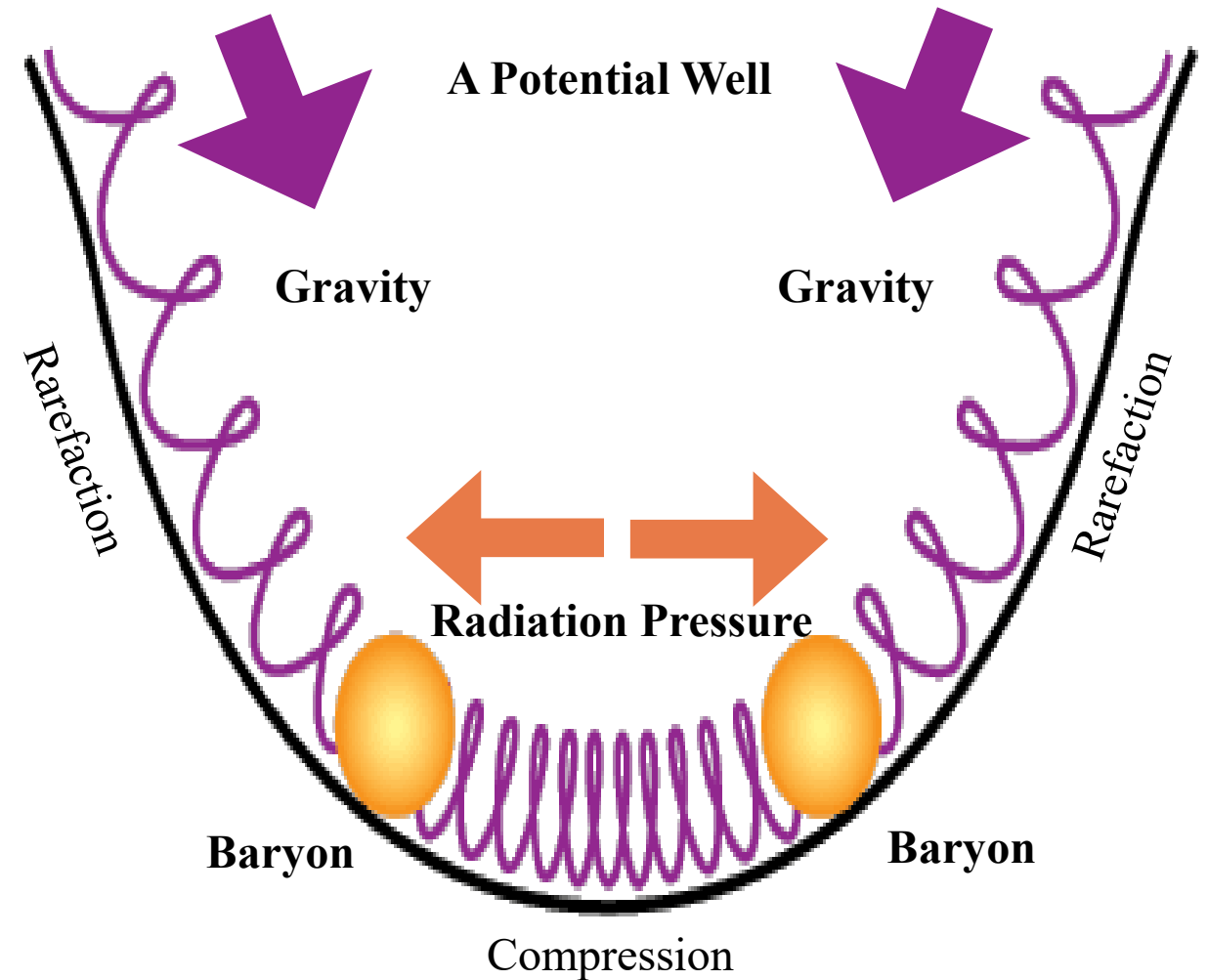


BARYON ACOUSTIC OSCILLATIONS (BAOs)

Image taken from but edited:

<http://background.uchicago.edu/~whu/beginners/sound.html>

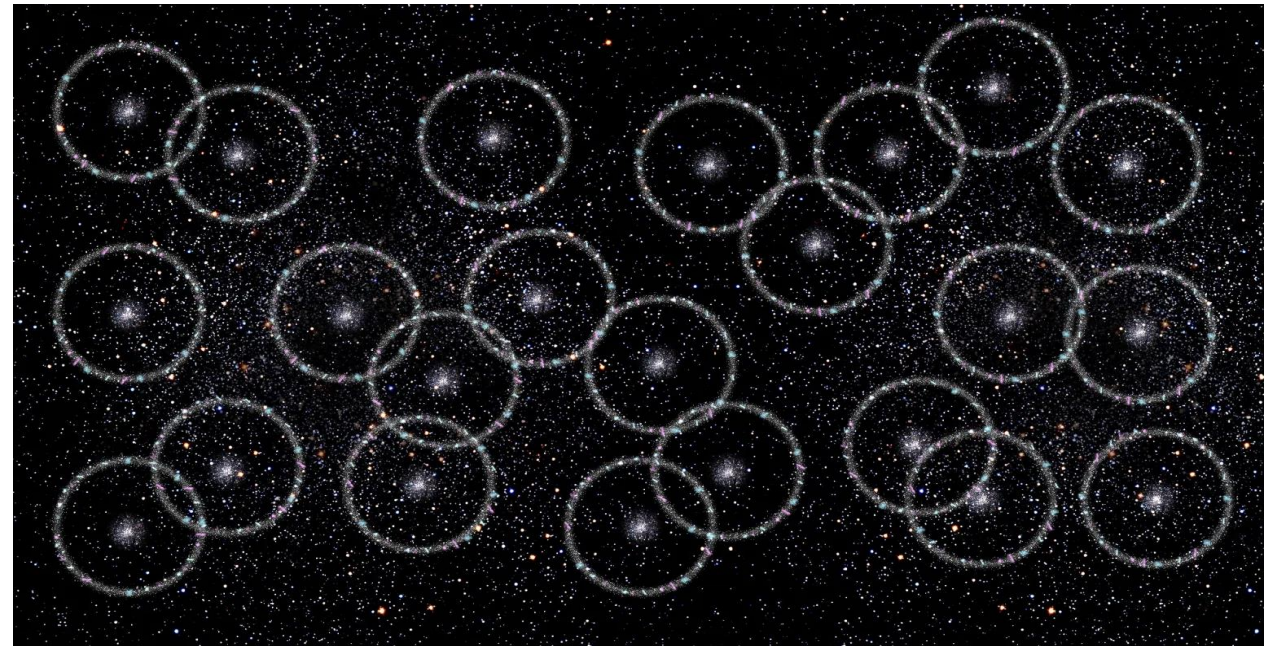
- In the early universe, baryons and photons existed as a fluid of plasma.
- Gravity compresses this fluid in the well, and the photons exert radiation pressure to resist.
- The system oscillates, with each force trying to outdo the other.
- Regions of compression and rarefactions can be seen, representing a sound wave.



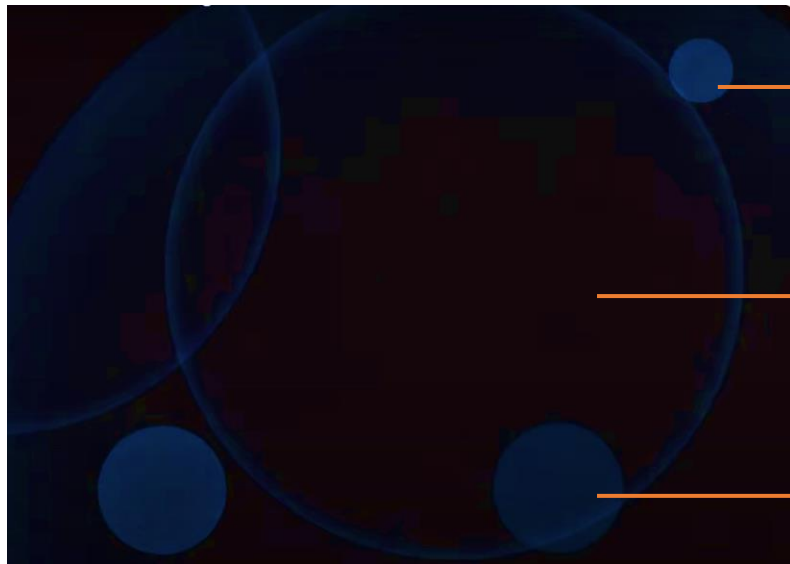
BARYON ACOUSTIC OSCILLATIONS



- Once photons decoupled from matter, the oscillations froze due to lack of radiation pressure.
- The size of anisotropy spots correspond to minimum, maximum or wavelengths between the two.
- This can help us figure out the curvature of the universe.



<https://bigthink.com/wp-content/uploads/2023/09/Baryon-Acoustic-Oscillation-v1-scaled-1.jpg?lb=1536,864>



Minimum wavelength

Maximum wavelength

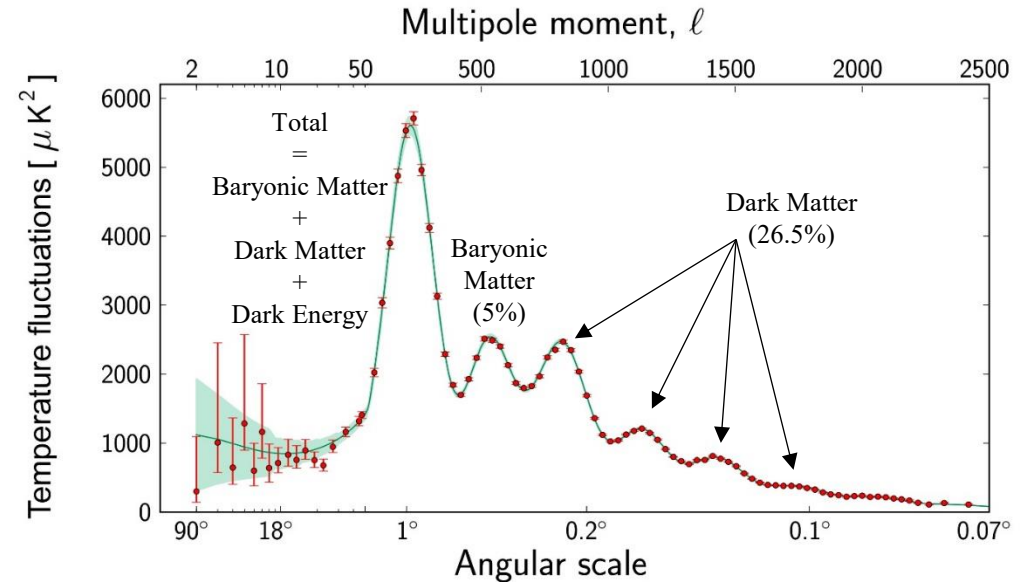
Anywhere in between

<https://www.youtube.com/watch?v=C4CKtEQJGM>
Timestamp: 5:35

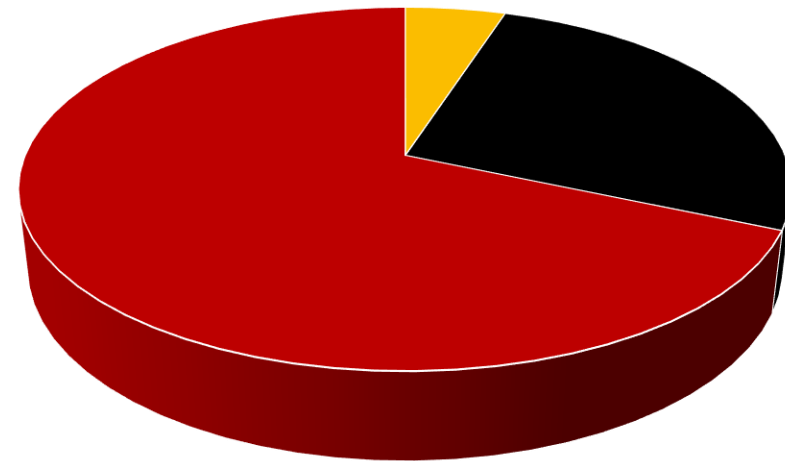


COMPOSITION OF THE UNIVERSE

- All possible sizes of anisotropy spots can be plotted on a Power Spectrum Diagram on the right.
 - The shape of the curve changes depending on the amount of each constituent of the Universe.
 - **First Peak:** Total Density of the Universe
 - **Second Peak:** Amount of Baryonic Matter (5%)
 - **Remaining Peaks:** Amount of Dark Matter (26.5%)
-
- **Baryonic Matter (Normal Matter):** Interacts with the EM spectrum and makes up all the galaxies and everything visible.
 - **Dark Matter:** Non-luminous matter that does not interact with the EM Spectrum but exerts gravitational attraction.
 - **Dark Energy:** Unknown force that accelerates the expansion of the Universe.



https://www.esa.int/ESA_Multimedia/Images/2013/03/Planck_Power_Spectrum



■ Baryonic Matter ■ Dark Matter ■ Dark Energy



CURVATURE OF THE UNIVERSE

Λ CDM – STANDARD COSMOLOGICAL MODEL

- A cosmological constant represents the curvature of the universe, Ω_k .
- In the standard lambda CDM model of the universe we take $\Omega_k = 0$, so that the universe is flat and there is no curvature.
- This is backed up by other data, such as baryonic acoustic oscillations.



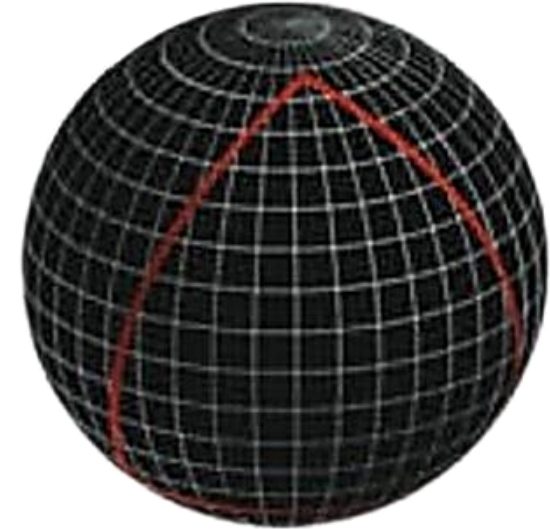
Λ CDM – GRAVITY AND DENSITY

- Gravity and the expansion rate of the universe will affect the shape of the universe.
- And hence as gravity and density are intrinsically linked, the shape of the universe is also linked to its own density.
- There are three types of geometry that the universe can take depending on these factors.



TYPES OF GEOMETRIES - SPHERICAL

- Density of the universe is great enough that gravity dominates over the universe's expansion.
- This is known as positive curvature, $\Omega_k > 0$.
- Geometry will be spherical – same as if it is on a sphere.
- Parallel lines will converge.
- Angles on a triangle drawn on a sphere add to more than 180° .

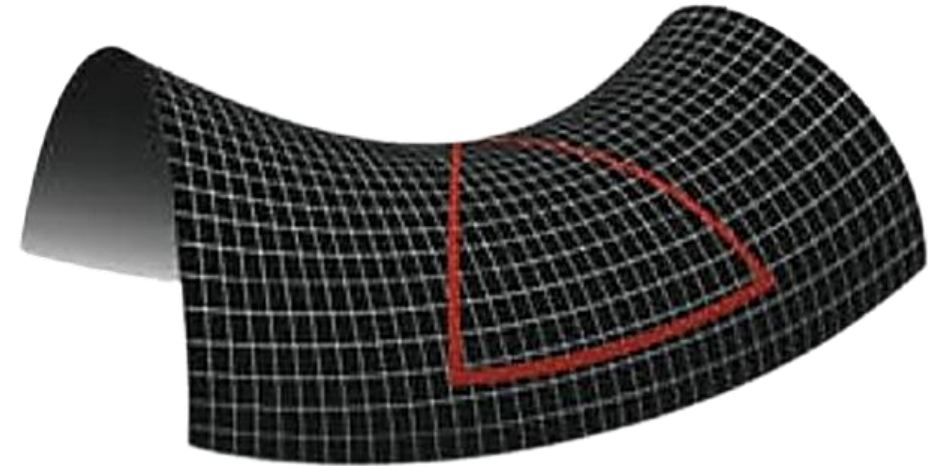


https://www.astronomy.com/wp-content/uploads/sites/2/2023/02/990006_320.jpg?resize=500%2C464



TYPES OF GEOMETRIES - ELLIPTICAL

- Density of universe too small so expansion of universe is dominant over gravity.
- Negative curvature, $\Omega_k < 0$.
- Geometry will be elliptical – same as if on a saddle-like shape.
- Parallel lines will diverge.
- Angles drawn on a triangle on an ellipse add to less than 180° .

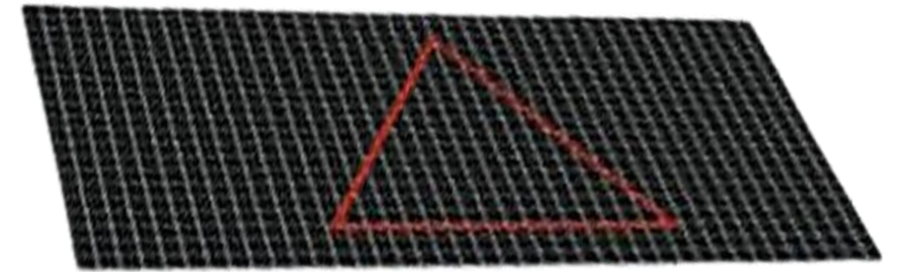


https://www.astronomy.com/wp-content/uploads/sites/2/2023/02/990006_320.jpg?resize=500%2C464



TYPES OF GEOMETRIES - FLAT

- Occurs at the critical density, the point where gravity and universal expansion balance out.
- Zero curvature, $\Omega_k = 0$.
- Flat geometry will be the same as geometry on a sheet of paper.
- Parallel lines stay parallel, angles on a triangle add to 180° .

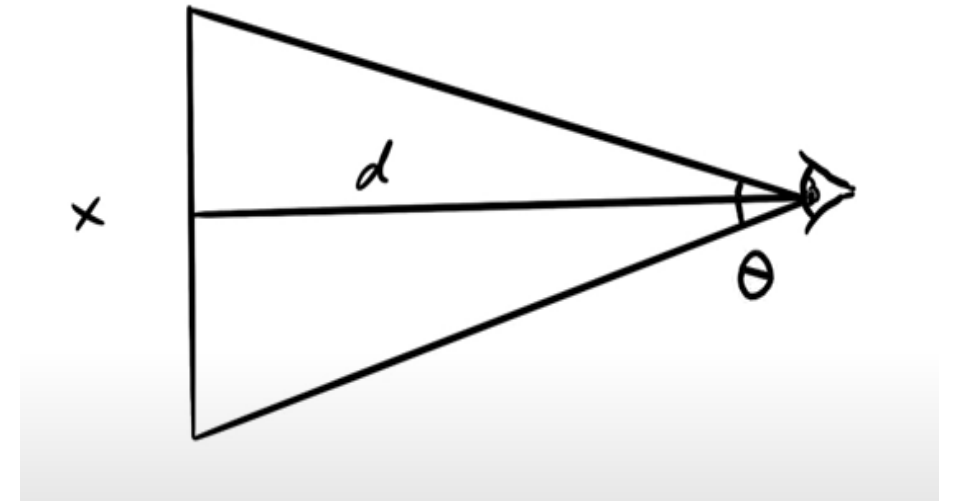


https://www.astronomy.com/wp-content/uploads/sites/2/2023/02/990006_320.jpg?resize=500%2C464



SO HOW CAN WE MEASURE THIS?

- Within an extremely small section of a sphere or a saddle geometry will seem to work as if we are on a flat plane.
- So, what we need is the largest possible triangle we can draw.
- The furthest distance we can observe is also the furthest back signal we can detect – the Cosmic Microwave Background Radiation – this is ‘d’.
- We need two distances known to form a triangle where we can measure angles. So, the second side is taken as the size of the anisotropies in the CMB data – this is ‘x’.
- But how can we measure the size of the anisotropies, x?



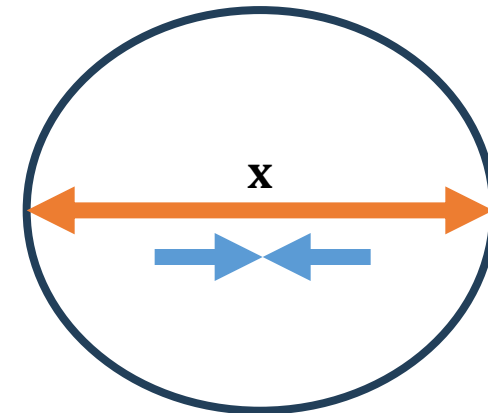
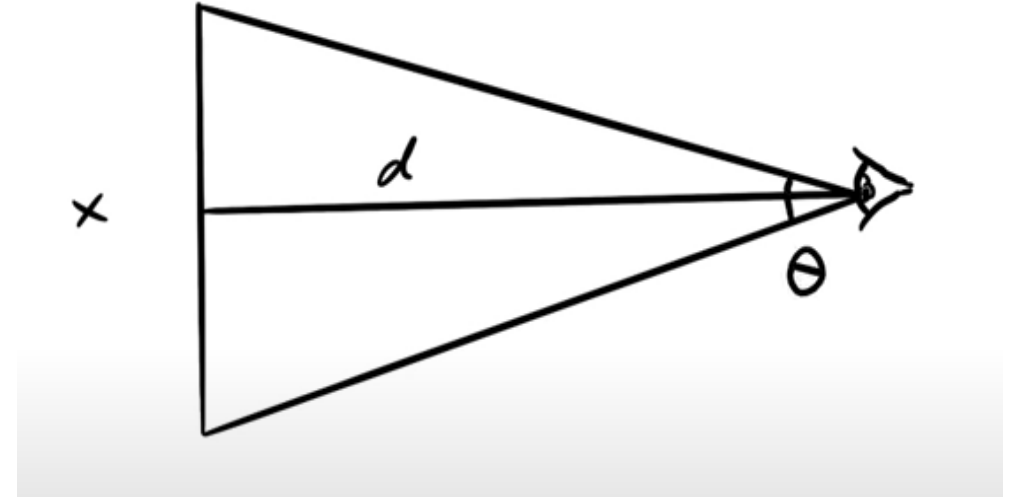
SO HOW CAN WE MEASURE THIS?

- We use the size of the largest anisotropy.
- This was formed when a BAO collapsed exactly once before recombination.
- This can be found by:

$$x = vt * \sigma$$

Where v is the speed of sound $= \frac{c}{2}$, $t = 380,000$ yrs and σ = expansion factor of the universe.

- Assuming that the universe is flat, we can use standard trigonometry to calculate the angle subtended by this distance on the CMB.
- This finds that the largest anisotropies should subtend an angle of 1° .

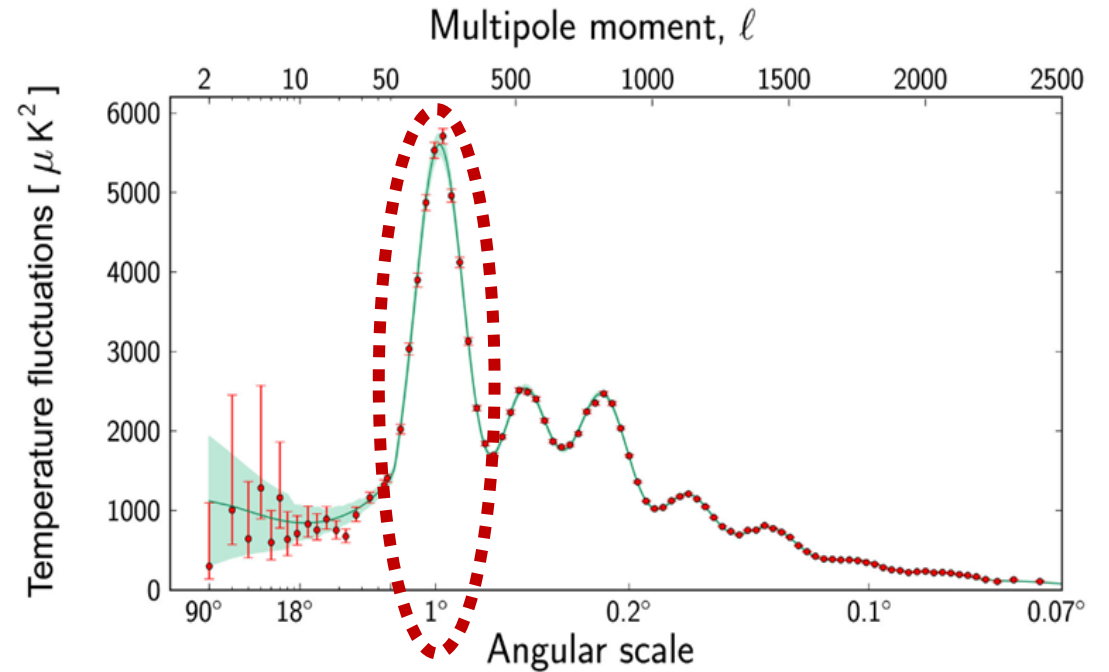


Collapsing BAO



SO HOW CAN WE MEASURE THIS?

- The power spectrum of the Planck data can be interpreted as a histogram of the frequency of the angles subtended by the anisotropies in the sky.
- The largest anisotropies occur at 1° , the peak highlighted on the power series.
- This is equal to what we predicted they should subtend – hence suggesting that the universe is flat.



THE RESULT

From here, we found

- $\Omega_k = 0.0007 \pm 0.0019$ - effectively flat.
- This agrees with the Λ CDM model where the universe is at the critical density and is flat.
- This also agrees with other data which is pointing that way too.



USING CMB TO FIND THE HUBBLE'S CONSTANT

WHAT IS HUBBLE'S CONSTANT?

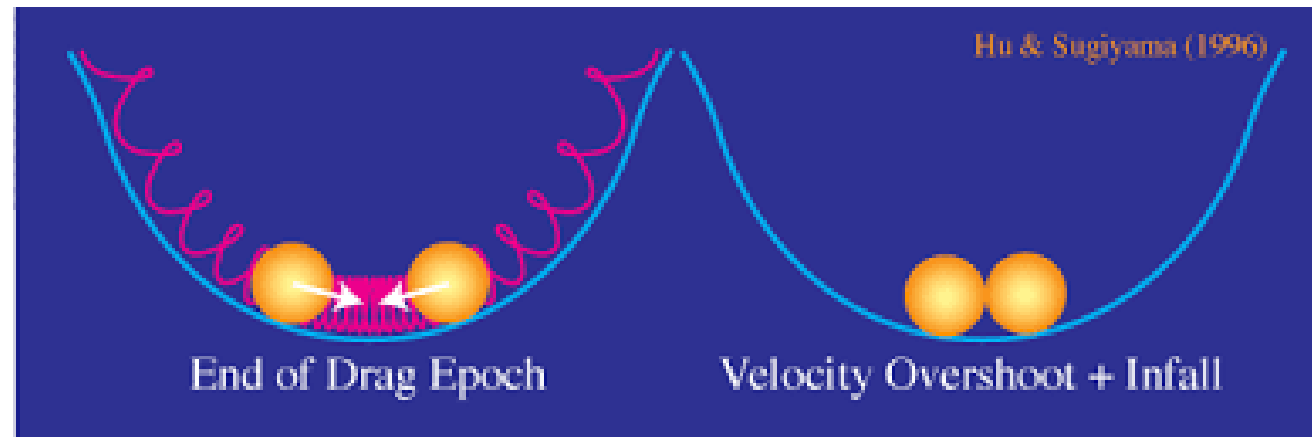
HOW DOES CMB RELATE TO IT?

- Hubble's constant can accurately determine the age of the universe.
- Currently the accepted value of Hubble's constant is $74.03 \pm 1.42 \text{ kms}^{-1} \text{Mpc}^{-1}$.
- CMB provides alternative methods to calculate a value of Hubble's constant.
- Methods involve baryonic acoustic oscillations and measuring redshift.



METHOD 1 – BARYONIC ACOUSTIC OSCILLATIONS

- Baryonic Acoustic oscillations, which are the sound waves emitted from the oscillation of protons and neutrons from the early universe.
- These are still visible upon the CMB map as they were still travelling until light could travel through the universe, which is when CMB was released.



<http://background.uchicago.edu/~whu/power/bao.html>



METHOD 1 – BARYONIC ACOUSTIC OSCILLATIONS

- The variations of baryonic density distribution, or baryonic acoustic oscillations, can be observed on CMB and can be used to obtain values of the density of objects in the early universe.
- From these densities a value of Hubble's constant can be found from:

$$\frac{H^2}{H_0^2} = \Omega_{0,R} a^{-4} + \Omega_{0,M} a^{-3} + \Omega_{0,k} a^{-2} + \Omega_{0,\Lambda}$$

- This method gives a value of the Hubble's constant as $67.4 \pm 0.50 \text{ km s}^{-1} \text{ Mpc}^{-1}$.



METHOD 2 – REDSHIFT

- CMB shows the history of redshifting radiation.
- The redshift of the early universe was very high ($z = 1100$), but currently the redshift is 0.
- CMB shows the redshift from the early universe, so the information provided by CMB can be used to find a value of Hubble's constant using the equation:

$$H_0 = \frac{cz}{d}$$



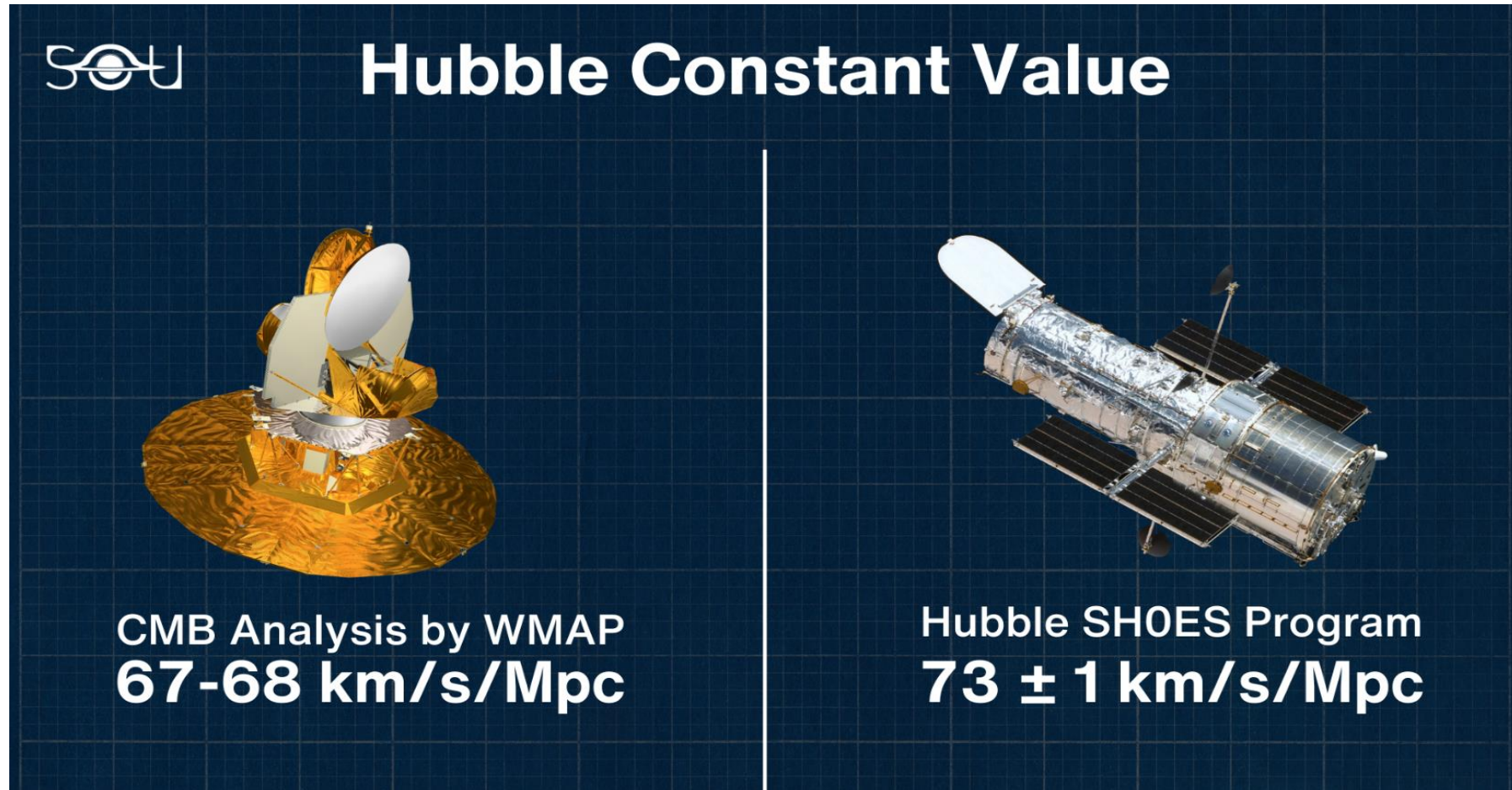
METHOD 2 – REDSHIFT

- Currently this method gives a value of H_0 as $67.80 \pm 0.77 \text{ kms}^{-1} \text{Mpc}^{-1}$.
- This method originally produced a value that was in agreement with the current accepted value of Hubble's constant.
- However, as the CMB map was measured in a higher level of detail the values of Hubble's constant began to decrease.



COMPARISON

The values of Hubble's constant obtained from both methods provided by CMB agree with each other, however both differ from the accepted value of $74.03 \pm 1.42 \text{ kms}^{-1} \text{Mpc}^{-1}$.



FURTHER RESEARCH & CONCLUSION

STEADY STATE vs BIG BANG

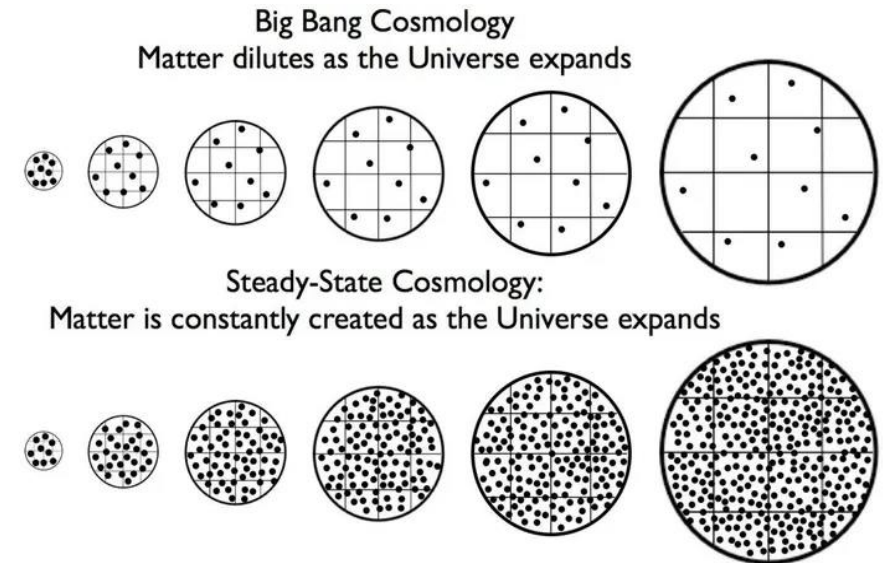
In the 20th century, there were 2 competing models of cosmological evolution: the steady state model and the big bang model.

Steady State:

- An infinitely old universe that is always the same in all locations. The universe is unchanging.
- As the universe expands, matter is constantly created. The average density of the universe remains constant.

Big Bang:

- The universe expanded from a single, dense point and the matter from this point dilutes as the universe expands.
- Opposes the steady state model as it suggests the universe had a beginning and can change over time.



<https://www.forbes.com/sites/startswithabang/2017/04/18/60-years-of-starstuff-how-humanity-discovered-where-our-elements-come-from/>

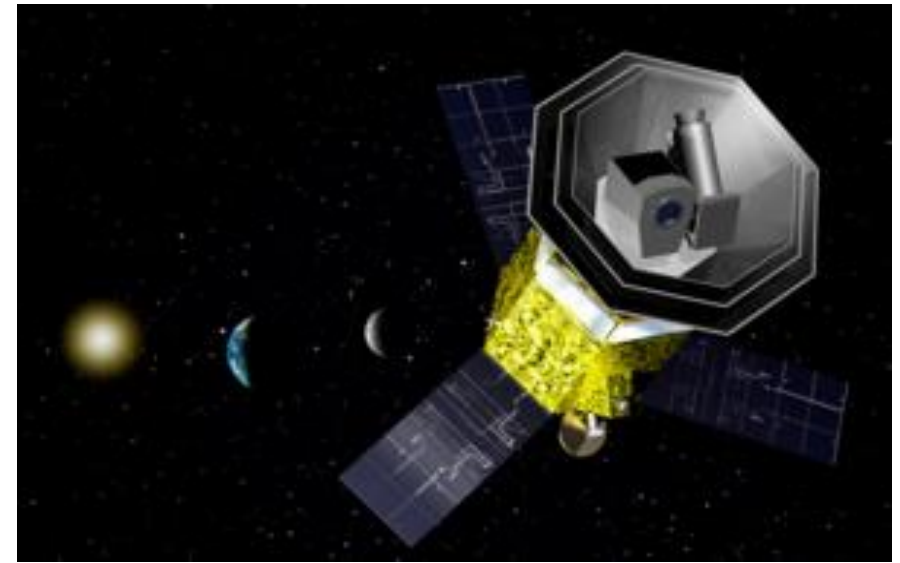


FUTURE RESEARCH

One target for cosmologists in the near future is to study the primordial B-modes - the curl component of the CMB polarisation. If satellites can differentiate between primordial B-modes and gravitational lensing, this could help enhance understanding of the inflation of the very early universe.

Example planned satellite missions:

- **LiteBIRD** - A satellite planned to be built and launched for 2028 by JAXA among other agencies
- **COrE+** - A planned ESA satellite
- **PIXIE** - A planned NASA satellite



Artist's Impression of The LiteBird Satellite
<https://www.gov.uk/government/case-studies/litebird>



CONCLUSION

- CMB is a vital discovery in cosmology providing evidence for the Big Bang model for the origin of the universe and opening the doors to many new discoveries such as a new value for Hubble's constant.
- The photons that make up the CMB are the remnants of particle interactions from soon after the Big Bang at a temperature of ~ 2.725 K with fluctuations at a difference of ~ 20 μ K caused by photons losing energy escaping gravitational potential wells of denser areas.
- Observations of the CMB have been made by the COBE, WMAP and Planck satellites. These observations confirmed the incredibly close match of the CMB to a blackbody spectrum and revealed the anisotropy of the temperature.



THANK YOU